MedPath India: Technical Design Document

AI-Powered Healthcare Navigator and Treatment Cost Estimator Platform

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1. Overview

1.1 Executive Summary

MedPath India is a real-time healthcare intelligence platform designed to help Indian patients—particularly those in Tier 2 and Tier 3 cities—discover optimal hospitals based on their medical condition, location, affordability, and personal context. The system translates patient intent expressed in natural language (Hindi, Hinglish, or English) into actionable healthcare recommendations with transparent cost estimates.

1.2 Problem Statement

Indian patients, especially in non-metro areas, face significant challenges:

- **Information asymmetry:** Difficulty finding appropriate hospitals for specific conditions
- **Cost uncertainty:** Unpredictable treatment costs and hidden charges
- **Language barriers:** Most health-tech solutions are English-first
- **Trust deficit:** No reliable way to assess hospital quality before visiting
- **Geographic constraints:** Limited awareness of accessible healthcare options

1.3 Solution Approach

MedPath India operates as a "Search-first + Intelligence-memory" architecture:

- **Real-time discovery:** Hospitals are discovered dynamically from public sources (Google Maps, Practo, JustDial, hospital websites, NABH records)
- **Intelligence memory:** The system stores learned intelligence (feedback, actual costs, trust signals) rather than maintaining a complete hospital database
- **Explainable recommendations:** Every recommendation includes transparent reasoning
- **Self-improving:** Feedback loops continuously improve ranking accuracy

1.4 Critical Disclaimer

This system is **decision-support only**. It does NOT:

- Diagnose diseases
- Replace medical professionals
- Provide medical advice
- Guarantee treatment outcomes

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2. Goals and Non-Goals

2.1 Goals

Goal	Success Metric	Priority
Enable patients to discover relevant hospitals within 60 seconds	Time-to-first-recommendation < 60s	P0
Provide cost estimates within 20% accuracy of actual costs	Cost estimation accuracy measured via feedback	P0
Support Hindi, Hinglish, and English natural language input	Intent extraction accuracy > 85%	P0
Deliver explainable recommendations	User understanding score > 4/5 in surveys	P0
Improve recommendations through feedback	Recommendation acceptance rate improvement over time	P1

Support Tier 2/3 city users effectively	Equal or better experience metrics vs metro users	P1
Maintain sub-3-second response times for recommendations	P95 latency < 3s	P1

2.2 Non-Goals

Non-Goal	Rationale
Medical diagnosis	Regulatory and safety concerns; out of scope
Appointment booking	Phase 2 feature; requires hospital integrations
Insurance claim processing	Separate product domain
Telemedicine/video consultations	Different product category
Maintaining complete India hospital database	Search-first architecture preferred
Real-time bed availability	Requires hospital system integrations
Prescription management	Out of scope for navigator product

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3. Product Architecture Overview

3.1 Core Architecture Philosophy

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│                         ARCHITECTURE PRINCIPLES
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──────┤
│ 1. Search-First: Discover hospitals dynamically, don't maintain full
DB      │
│ 2. Intelligence-Memory: Store learnings, not raw hospital data
│
│ 3. Explainability-First: Every recommendation must be justifiable
│
│ 4. Graceful Degradation: System works even when external sources fail
│
│ 5. Privacy-by-Design: Minimize PII storage, anonymize feedback
│
│ 6. Modular Monolith: Start simple, extract services when needed
│
└─────────────────────────────────────────────────────────────────────
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3.2 System Boundaries

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│                              SYSTEM BOUNDARY
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│
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┌─────────────────────────────────────────────────┐   │
│  │   PATIENT    │──▶│           MEDPATH INDIA PLATFORM
│   │
│  │   (User)     │◀──│
│   │
│  └──────────────┘
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│                      ┌─────────────────────────────────────┐
│
│                      │      EXTERNAL DATA SOURCES          │
│
│                      │  (Google Maps, Practo, JustDial,    │
│
│                      │   NABH, Hospital Websites)          │
│
│                      └─────────────────────────────────────┘
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│
│
│  OUT OF SCOPE: Hospital internal systems, Insurance systems,
│
│                Pharmacy systems, Lab systems
│
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```

4. End-to-End System Workflow

4.1 Primary User Flow

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│                        END-TO-END USER JOURNEY
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PHASE 1: INPUT CAPTURE
┌──────────────────────────────────────────────────────────────────────
──────┐
│ Step 1: User opens app
│
│ Step 2: User provides input (voice OR text)
│
│ Step 3: If voice → Speech-to-Text conversion
│
│ Step 4: Raw text captured in Hindi/Hinglish/English
│
└──────────────────────────────────────────────────────────────────────
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                                   │
                                   ▼
PHASE 2: INTENT EXTRACTION
┌──────────────────────────────────────────────────────────────────────
──────┐
│ Step 5: NLU pipeline processes text
│
│ Step 6: Extract:
│
│         - Symptoms/medical condition
│
│         - Possible medical category
│
│         - Urgency level
│
│         - Patient context (if provided)
│
```

```
│ Step 7: Identify missing MANDATORY information
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└───────────────────────────────────────────────────────────────────────
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                                    │
                                    ▼
PHASE 3: CONTEXT COMPLETION
┌───────────────────────────────────────────────────────────────────────
──────┐
│ Step 8: Check mandatory context:
│
│         - Location (GPS or address) ← MANDATORY
│
│         - Medical issue/symptoms ← MANDATORY
│
│ Step 9: If missing → Generate follow-up questions
│
│ Step 10: Collect optional context if volunteered:
│
│          - Age, comorbidities, insurance, budget, hospital preference
│
└───────────────────────────────────────────────────────────────────────
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                                    │
                                    ▼
PHASE 4: HOSPITAL DISCOVERY
┌───────────────────────────────────────────────────────────────────────
──────┐
│ Step 11: Determine search radius based on:
│
│          - Urgency (emergency = smaller radius)
│
│          - City density (metro vs rural)
│
│          - Speciality rarity (rare = wider radius)
│
│ Step 12: Query external sources in parallel:
```

```
│
│           - Google Maps API
│
│           - Practo scraper/API
│
│           - JustDial scraper
│
│           - NABH database lookup
│
│ Step 13: Aggregate and deduplicate results
│
└────────────────────────────────────────────────────────────────────────
──────┘
                                    │
                                    ▼
PHASE 5: FILTERING
┌────────────────────────────────────────────────────────────────────────
──────┐
│ Step 14: Apply exclusion filters:
│
│           - Irrelevant speciality
│
│           - Poor review sentiment (< threshold)
│
│           - Extreme distance
│
│           - Weak emergency capability (if urgent)
│
│           - Affordability mismatch (if budget provided)
│
│           - Low trust score (from intelligence memory)
│
└────────────────────────────────────────────────────────────────────────
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                                    │
                                    ▼
PHASE 6: ENRICHMENT
```

```
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───────┐
│ Step 15: Enrich filtered hospitals with intelligence memory:
│
│          - Past feedback scores
│
│          - Real treatment costs (from previous users)
│
│          - Hidden charge warnings
│
│          - Wait time estimates
│
│          - Trust signals
│
│ Step 16: Fetch real-time data where possible:
│
│          - Current reviews
│
│          - Operating hours
│
│          - Contact information
│
└──────────────────────────────────────────────────────────────────────
───────┘
                                    │
                                    ▼
PHASE 7: RANKING
┌──────────────────────────────────────────────────────────────────────
───────┐
│ Step 17: Calculate composite score using weighted factors:
│
│          HIGH PRIORITY:
│
│          - Medical specialization relevance (weight: 25%)
│
│          - Hospital capability match (weight: 20%)
│
```

```
│           - Distance/travel time (weight: 15%)
│
│           - Affordability score (weight: 15%)
│
│           - Review sentiment (weight: 10%)
│
│           - Accreditation (weight: 10%)
│
│           - Historical feedback (weight: 5%)
│
│           MEDIUM PRIORITY (tie-breakers):
│
│           - Wait times, billing transparency, suitability flags
│
│ Step 18: Generate explanation for each rank
│
└───────────────────────────────────────────────────────────────────────
──────┘
                                    │
                                    ▼
PHASE 8: COST ESTIMATION
┌───────────────────────────────────────────────────────────────────────
──────┐
│ Step 19: For top-ranked hospitals, generate cost estimates:
│
│           - Procedure cost
│
│           - Consultation fees
│
│           - Diagnostics
│
│           - Medicines (estimated)
│
│           - Hospital stay (if applicable)
│
│           - Contingency buffer
│
```

```
│ Step 20: Apply adjustments for:
│
│          - City tier
│
│          - Hospital type (govt/private/trust)
│
│          - Patient comorbidities
│
│          - Historical actuals from feedback
│
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──────┘
                                   │
                                   ▼
PHASE 9: PRESENTATION
┌──────────────────────────────────────────────────────────────────────
──────┐
│ Step 21: Render results as hospital cards:
│
│          - Name, ratings, distance, travel time
│
│          - Estimated cost range
│
│          - Key facilities
│
│          - "Why recommended" explanation
│
│          - Trust score, accreditation badges
│
│          - Action buttons (call, navigate)
│
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PHASE 10: POST-TREATMENT FEEDBACK
┌──────────────────────────────────────────────────────────────────────
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───────┐
│ Step 22: After treatment (triggered via notification):
│
│          - Actual treatment cost
│
│          - Hospital experience rating
│
│          - Doctor experience rating
│
│          - Hidden charges encountered
│
│          - Wait time experienced
│
│          - Billing transparency rating
│
│          - Recommendation likelihood (NPS)
│
│ Step 23: Update intelligence memory
│
│ Step 24: Retrain/adjust ranking weights periodically
│
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```

5. High-Level Architecture Diagram

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┌──────────────────────────────────────────────────────────────────────────
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│                                    MEDPATH INDIA - HIGH LEVEL
ARCHITECTURE                    │
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                                          ┌─────────────────┐
                                          │      WEB APP    │
                                          │                 │
                                          └────────┬────────┘
                                                   │
                                                   │ HTTPS
                                                   ▼
┌──────────────────────────────────────────────────────────────────────────
──────────────────────┐
│                                       API GATEWAY / LOAD BALANCER
│
│                                        (nginx / AWS ALB)
│
└────────────────────────────────────────────────────┬─────────────────────
──────────────────────┘
                                                   │
┌─────────────────────────────┼─────────────────────────────┐
                    │                             │
│
                    ▼                             ▼
▼
         ┌──────────────────┐          ┌──────────────────┐
┌──────────────────┐
         │   VOICE SERVICE  │          │   CORE API       │          │
FEEDBACK API    │
         │  (Speech-to-Text)│          │   SERVICE        │          │
SERVICE         │
         └────────┬─────────┘          └────────┬─────────┘
└────────┬─────────┘
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                    ┌─────────────────────┐
┌──────────────────────────┐
                    │   CONVERSATIONAL    │       │   HOSPITAL DISCOVERY
│
                    │   AI ENGINE         │       │   ORCHESTRATOR
│
                    │                     │       │
│
                    │ - NLU Pipeline      │       │ - Source Aggregator
│
                    │ - Intent Extraction │       │ - Filter Engine
│
                    │ - Follow-up Engine  │       │ - Ranking Engine
│
                    │ - Context Manager   │       │ - Cost Estimator
│
                    └──────────┬──────────┘
└────────────┬─────────────┘
                              │                                │
                              │                                │
              ┌───────────────┴───────────────┐                │
              │                               │                │
              ▼                               ▼                ▼
    ┌─────────────────┐             ┌─────────────────┐
┌──────────────────────┐
    │   LLM SERVICE   │             │  LOCATION       │  │  EXTERNAL
SOURCE    │
    │   (OpenAI /     │             │  INTELLIGENCE   │  │  ADAPTERS
│
```

```
      │    Anthropic)   │            │  SERVICE        │  │
│
      └─────────────────┘            └─────────────────┘  │  - Google Maps
│
                                                          │  - Practo
│
                                                          │  - JustDial
│
                                                          │  - NABH
│
└──────────┬──────────┘
                                                                   │
                                                                   │
HTTP/Scraping
                                                                   ▼
┌──────────────────────┐
                                                          │  EXTERNAL
SOURCES   │
                                                          │  (Internet)
│
└──────────────────────┘
┌────────────────────────────────────────────────────────────────────────
──────────────────────┐
│                                        DATA LAYER
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│  ┌─────────────────┐     ┌─────────────────┐     ┌─────────────────┐
┌─────────────────┐   │
│  │   PostgreSQL    │     │     Redis       │     │  Elasticsearch  │
│   S3 / Blob     │   │
```

```
│  │                 │    │                 │    │  (Optional)     │
│   Storage       │   │
│  │ - User sessions │    │ - Session cache │    │                 │
│                 │   │
│  │ - Feedback data │    │ - Query cache   │    │ - Hospital      │
│ - Audit logs    │   │
│  │ - Hospital      │    │ - Rate limiting │    │   intelligence  │
│ - ML artifacts  │   │
│  │   intelligence  │    │ - Hospital      │    │   search        │
│                 │   │
│  │ - Cost history  │    │   cache         │    │                 │
│                 │   │
│  │ - Audit logs    │    │                 │    │                 │
│                 │   │
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└─────────────────┘   │
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```

6. Low-Level Component Architecture

6.1 Component Breakdown

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──────────────────────┐
│                           COMPONENT ARCHITECTURE DETAIL
│
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┌─────────────────────────────────────────────────────────────────────
──────────────────────┐
│                                 PRESENTATION LAYER
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│
│
┌─────────────────────────────────────────────────────────────────────
──────────────┐    │
│  │                             MOBILE APPLICATION
│    │
│  │
│    │
│  │  ┌─────────────┐  ┌─────────────┐  ┌─────────────┐  ┌─────────────┐
┌────────────┐  │    │
│  │  │   Voice     │  │   Chat      │  │  Results    │  │  Hospital   │
│ Feedback  │  │    │
│  │  │   Input     │  │   Interface │  │  Display    │  │  Detail     │
│ Form       │  │    │
│  │  │   Screen    │  │   Screen    │  │  Screen     │  │  Screen     │
│ Screen     │  │    │
│  │  └─────────────┘  └─────────────┘  └─────────────┘  └─────────────┘
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──────┐    │    │
```

```
│  │  │                          SHARED COMPONENTS
│    │    │
│  │  │  - VoiceRecorder    - MessageBubble    - HospitalCard    -
CostBreakdown    │    │    │
│  │  │  - LocationPicker   - LoadingStates    - RankExplainer   -
FeedbackStars    │    │    │
│  │
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┌─────────────────────────────────────────────────────────────────────────
──────────────────────┐
│                                    APPLICATION LAYER
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──────────────────────┤
│
│
│  ┌────────────────────────────────────────┐
┌────────────────────────────────────────┐       │
│  │          CORE API SERVICE             │  │       VOICE SERVICE
│       │
│  │                                       │  │
│       │
│  │  ┌──────────────────────────────────┐  │  │
┌──────────────────────────────────┐  │       │
│  │  │     ConversationController      │  │  │  │
SpeechToTextController      │  │       │
│  │  │     - POST /conversation        │  │  │  │     - POST
/voice/transcribe    │  │       │
```

```
│  │  │     - GET /conversation/{id}    │  │  │
└──────────────────────────────────┘  │       │
│  │  └──────────────────────────────────┘  │  │                  │
│       │
│  │                                        │  │                  ▼
│       │
│  │  ┌──────────────────────────────────┐  │  │
┌──────────────────────────────────┐  │       │
│  │  │     RecommendationController    │  │  │  │
SpeechToTextService         │  │       │
│  │  │     - GET /recommendations/{id} │  │  │  │     - Google Speech
API          │  │       │
│  │  └──────────────────────────────────┘  │  │  │     - Language
detection        │  │       │
│  │                                        │  │  │     - Hinglish
handling         │  │       │
│  │  ┌──────────────────────────────────┐  │  │
└──────────────────────────────────┘  │       │
│  │  │     FeedbackController          │  │  │
│       │
│  │  │     - POST /feedback            │  │
└──────────────────────────────────────┘       │
│  │  │     - GET /feedback/pending     │  │
│
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│  │                                        │
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```
│                                    DOMAIN LAYER
│
├────────────────────────────────────────────────────────────────────────
──────────────────────┤
│
│
│
┌────────────────────────────────────────────────────────────────────────
───────────────┐    │
│  │                           CONVERSATIONAL AI ENGINE
│    │
│  │
│    │
│  │  ┌──────────────────┐  ┌──────────────────┐  ┌──────────────────┐
┌──────────────────┐ │    │
│  │  │ NLUProcessor     │  │ IntentExtractor │  │ ContextManager  │  │
FollowUpEngine  │ │    │
│  │  │                  │  │                  │  │                  │  │
│ │    │
│  │  │ - Tokenization  │  │ - Symptom        │  │ - Session state │  │ -
Missing field │ │    │
│  │  │ - Language      │  │   extraction    │  │ - Context       │  │
detection     │ │    │
│  │  │   detection     │  │ - Urgency        │  │   accumulation  │  │ -
Question      │ │    │
│  │  │ - Normalization │  │   assessment    │  │ - Validation    │  │
generation    │ │    │
│  │  └──────────────────┘  │ - Category      │  └──────────────────┘
└──────────────────┘ │    │
│  │                        │   mapping       │
│    │
│  │                        └──────────────────┘
│    │
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───────────────┘    │
│
```

```
│
│
┌────────────────────────────────────────────────────────────────────────
──────────────┐    │
│  │                         HOSPITAL DISCOVERY ORCHESTRATOR
│    │
│  │
│    │
│  │  ┌─────────────────┐  ┌─────────────────┐  ┌─────────────────┐
┌─────────────────┐ │    │
│  │  │ SourceAggregator│  │ FilterEngine    │  │ RankingEngine   │  │
CostEstimator   │ │    │
│  │  │                 │  │                 │  │                 │  │
│ │    │
│  │  │ - Parallel      │  │ - Speciality    │  │ - Weighted      │  │ -
Component     │ │    │
│  │  │   fetching      │  │   filter        │  │   scoring       │  │
breakdown     │ │    │
│  │  │ - Deduplication │  │ - Distance      │  │ - Explanation   │  │ -
City tier     │ │    │
│  │  │ - Normalization │  │   filter        │  │   generation    │  │
adjustment    │ │    │
│  │  │ - Timeout       │  │ - Trust filter  │  │ - Tie-breaking  │  │ -
Comorbidity   │ │    │
│  │  │   handling      │  │ - Budget filter │  │                 │  │
adjustment    │ │    │
│  │  └─────────────────┘  └─────────────────┘  └─────────────────┘
└─────────────────┘ │    │
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│
│
┌────────────────────────────────────────────────────────────────────────
──────────────┐    │
│  │                           LOCATION INTELLIGENCE SERVICE
```

```
│    │
│  │
│    │
│  │  ┌──────────────────┐  ┌──────────────────┐  ┌──────────────────┐
│    │
│  │  │ GeocodingService│  │ RadiusCalculator│  │ TravelTimeCalc  │
│    │
│  │  │                 │  │                 │  │                 │
│    │
│  │  │ - Address to    │  │ - Urgency-based │  │ - Distance      │
│    │
│  │  │   coordinates   │  │   radius        │  │   matrix        │
│    │
│  │  │ - GPS fallback  │  │ - City density  │  │ - Traffic       │
│    │
│  │  │ - Validation    │  │   adjustment    │  │   estimation    │
│    │
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